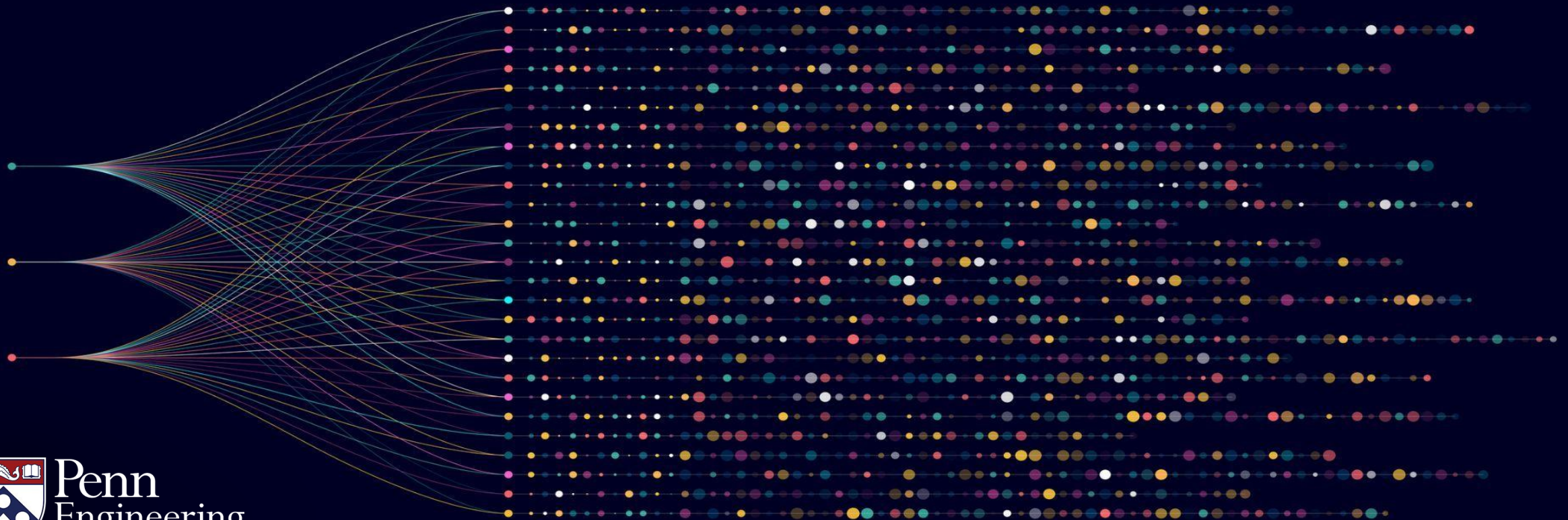


Algorithms for Big Data

Anindya De

Erik Waingarten

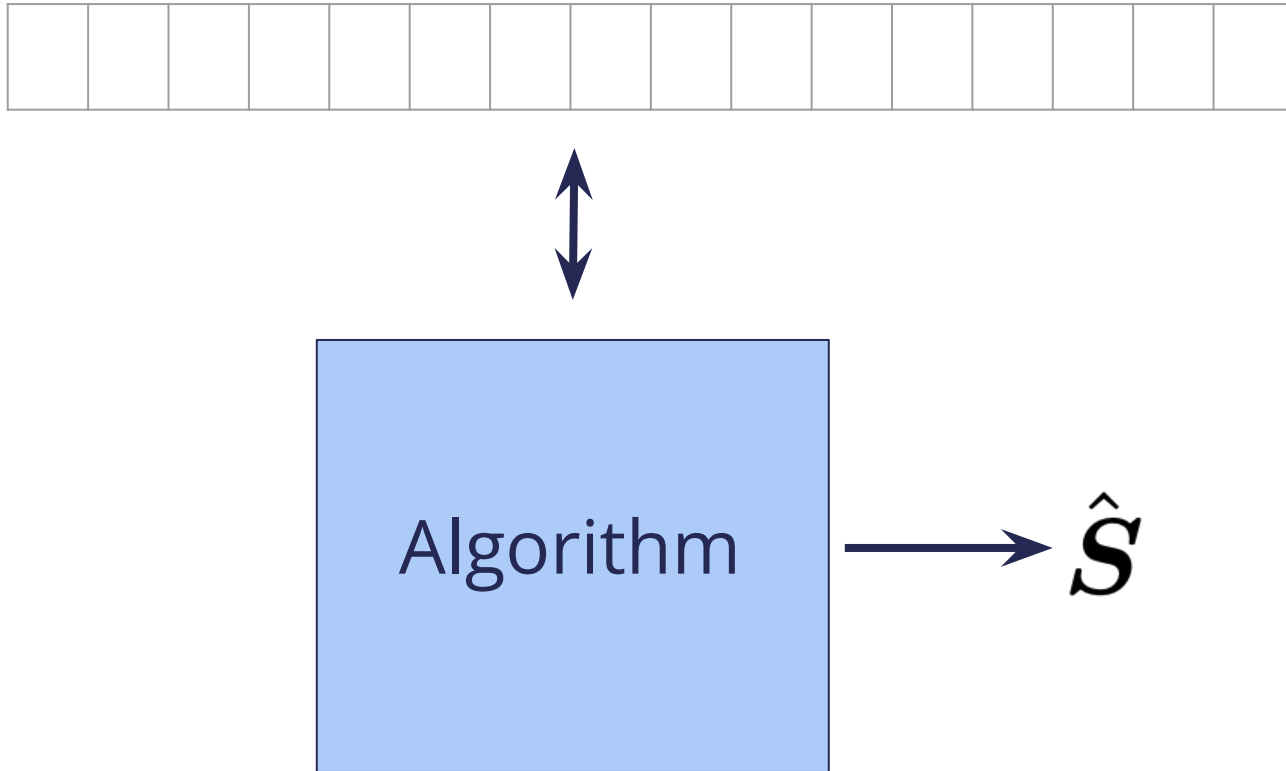
Description of the Algorithm



Objectives: Computing Large Averages

- Algorithm for approximating large averages
- Questions for the analysis

Algorithm Description



1. Sample q positions:

$$\mathbf{i}_1, \dots, \mathbf{i}_q \sim [n]$$

1. Output average:

$$\hat{\mathbf{S}} = \frac{1}{q} \sum_{t=1}^q X_{\mathbf{i}_t}$$

Understanding the algorithm:

- Queries are each chosen uniformly and independently

- For every $j \in [n]$,

$$\Pr_{\mathbf{i}_t \sim [n]} [\mathbf{i}_t = j] = \frac{1}{n}$$

1. Sample q positions:

$$\mathbf{i}_1, \dots, \mathbf{i}_q \sim [n]$$

1. Output average:

$$\hat{\mathbf{S}} = \frac{1}{q} \sum_{t=1}^q X_{\mathbf{i}_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$i_1 =$

$i_2 =$

$\vdots$

$i_q =$

source of randomness:



1. Sample q positions:

$$i_1, \dots, i_q \sim [n]$$

1. Output average:

$$\hat{S} = \frac{1}{q} \sum_{t=1}^q X_{i_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$i_1 = 11$$

$$i_2 = 8$$

$\vdots$

$$i_q = 15$$

source of randomness:



1. Sample q positions:

$$i_1, \dots, i_q \sim [n]$$

1. Output average:

$$\hat{S} = \frac{1}{q} \sum_{t=1}^q X_{i_t}$$

Example Execution of Algorithm:

0	0	1	1	1	1	0	1	0	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$i_1 = 11$$

$$i_2 = 8$$

$\vdots$

$$i_q = 15$$

source of randomness:



1. Sample q positions:

$$i_1, \dots, i_q \sim [n]$$

1. Output average:

$$\hat{S} = \frac{1}{q} \sum_{t=1}^q X_{i_t}$$

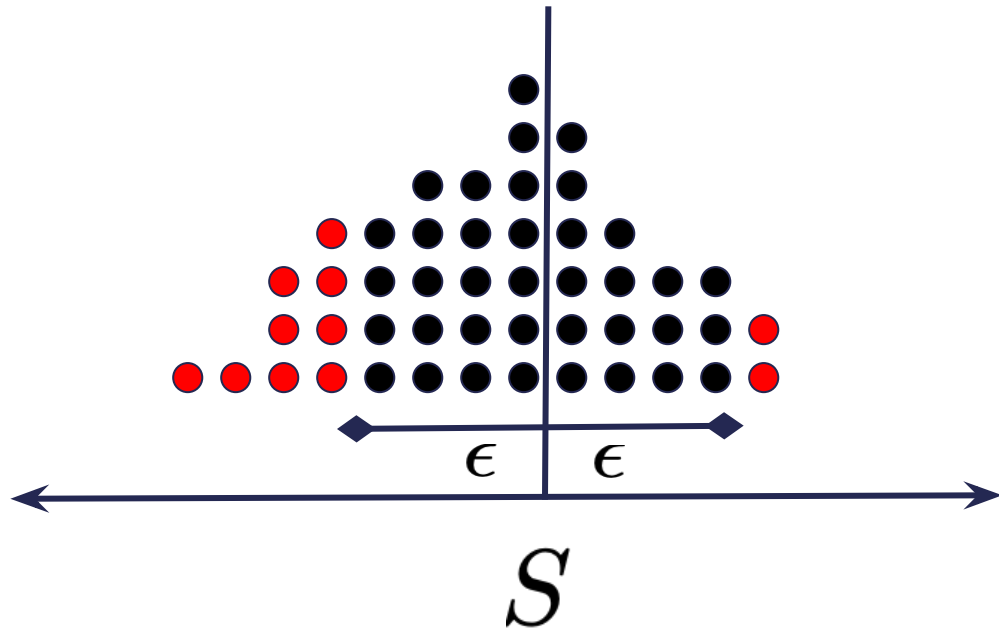
Questions for the Algorithm Analysis:

- How many queries do we need?

$$\Pr_{\mathbf{i}_1, \dots, \mathbf{i}_q \sim [n]} \left[\left| \hat{\mathbf{S}} - \mathcal{S} \right| \leq \epsilon \right] \geq 1 - \delta.$$

- Should depend on ϵ, δ .

Questions for the Algorithm Analysis:



$$\hat{S} = \frac{1}{q} \sum_{t=1}^q X_{i_t}$$

$$\Pr_{i_1, \dots, i_q \sim [n]} \left[\left| \hat{S} - S \right| \leq \epsilon \right] \geq 1 - \delta.$$

ϵ : accuracy δ : failure probability

Toward the Analysis:

- If we took infinitely many queries, would the algorithm's output approach the average?
 - Hint: think of the expectation and the central-limit theorem!
- Suppose yes, with infinitely many queries. What happens with finitely many queries?
 - How close is an outcome to its expectation?
 - Hint: think of the variance and Chebyshev's inequality